



2021 HSC Course Assessment Task 3 - Trial

18 June 2021

- Working time – 3 hours. (plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.

- Mark your answers on the answer sheet provided (numbered as page 13)

- Commence each new section on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete writing.

Class (please tick)

○ 12MM4C Mr Berry

[illegible]

Section I (10 marks)

Attempt Questions 1 to 10

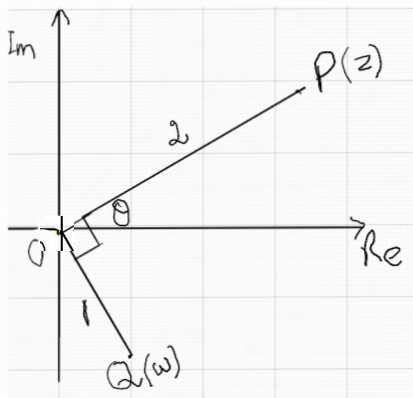
Allow approximately 15 minutes for this section.

1. A simplified form of $\left(\frac{\cos 2\theta + i \sin 2\theta}{\cos \theta - i \sin \theta} \right)$ is
- A. $\cos \theta + i \sin \theta$
 - B. $\cos 3\theta + i \sin 3\theta$
 - C. $\cos \theta - i \sin \theta$
 - D. $\cos 3\theta - i \sin 3\theta$
2. The contrapositive statement to "If a function is even then its inverse is not even" is:
- A. If the inverse is not even then the function is even.
 - B. If the inverse is not even then the function is not even.
 - C. If the inverse is even then then the function is even.
 - D. If the inverse is even then the function is not even.
3. Which of the following expressions is the partial fraction form of the algebraic fraction $\frac{x-1}{(x-2)^2(x^2+3)}$?
- P, Q, R and S are constants,
- A. $\frac{P}{(x-2)^2} + \frac{Q}{x^2+3}$.
 - B. $\frac{P}{(x-2)^2} + \frac{Qx+R}{x^2+3}$.
 - C. $\frac{P}{(x-2)} + \frac{Q}{(x-2)^2} + \frac{R}{x^2+3}$.
 - D. $\frac{P}{(x-2)} + \frac{Q}{(x-2)^2} + \frac{Rx+S}{x^2+3}$.

4. z is a complex number such that $z = (1 - a \cos \theta) + i(1 - b \sin \theta)$, where a, b and θ are all real. If the locus of z in an Argand diagram is the straight line $x + y = 1$, which of the following is true?

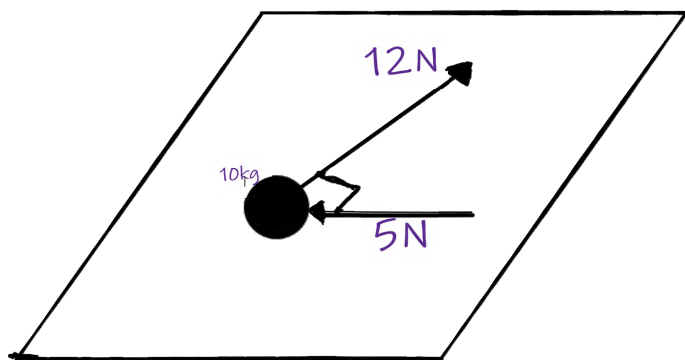
- A. $a^2 + b^2 \leq 1$.
- B. a and b are free to take any real values.
- C. $a^2 + b^2$ must be equal to 1.
- D. The values of a and b are not restricted and there is no relationship between a and b .

5. In the Argand diagram below. P and Q represent the complex numbers z and ω respectively. $\angle POQ = \frac{\pi}{2}$, $|z| = 2$ units and $|\omega| = 1$ unit. Which of the following is correct?



- A. $\omega = -e^{i(\frac{\pi}{2}-\theta)}$
- B. $\omega = e^{-i(\frac{\pi}{2}-\theta)}$
- C. $\omega = -e^{-i(\frac{\pi}{2}-\theta)}$
- D. $\omega = e^{i(\frac{\pi}{2}-\theta)}$

6. A 10 kg object placed on a smooth horizontal table is subject to two horizontal forces of 12N and 5N in two perpendicular directions as shown.



Calculate the acceleration of the object.

- A. $1.7ms^{-2}$
- B. $13ms^{-2}$
- C. $1.3ms^{-2}$
- D. $0.7ms^{-2}$

7. Using a suitable substitution, $\int_1^{\sqrt{3}} \frac{\log(\tan^{-1} x)}{1+x^2} dx$ can be expressed in terms of u as:

- A. $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \log u \, du$
- B. $\int_0^{\frac{\pi}{3}} \log u \, du$
- C. $\int_0^{\frac{\pi}{2}} \frac{\log(u)}{1+\tan^2 u} \, du$
- D. $\int_0^{\frac{\pi}{6}} \frac{\log(u)}{1+\tan^2 u} \, du$

8. Let $z = \left(\frac{1+i}{\sqrt{2}}\right)^i$ and $\omega = \left(\frac{1-i}{\sqrt{2}}\right)^i$. Which of the following statement is true about the relationship between z and ω ?

- A. $z < \omega$.
- B. $z > \omega$.
- C. complex numbers cannot be ordered. Therefore neither $z < \omega$ nor $z > \omega$ can be true.
- D. $|z| = |\omega|$.

9. The polynomial $P(z)$ has the equation $P(z) = z^4 - 2z^3 + \alpha z + 15$, where α is real. Given that $(2 - i)$ is a zero of $P(z)$, which of the following expressions is $P(z)$ as a product of two real quadratic factors?

A. $(z^2 + 4z + 5)(z^2 + 2z + 3)$

B. $(z^2 - 4z + 5)(z^2 + 2z + 3)$

C. $(z^2 + 4z + 5)(z^2 - 2z + 3)$

D. $(z^2 - 4z + 5)(z^2 - 2z + 3)$

10. The equations of the lines l_1 , l_2 and l_3 are give by

$l_1: \mathbf{r} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k} + \lambda(2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k})$

$l_2: \mathbf{r} = 2\mathbf{i} - 5\mathbf{j} + 4\mathbf{k} + \mu(\mathbf{i} + \mathbf{j} + \mathbf{k})$

$l_3: \mathbf{r} = -3\mathbf{i} + 2\mathbf{j} - \mathbf{k} + \nu(\mathbf{i} + \mathbf{j} + \frac{1}{5}\mathbf{k})$

where λ , μ and ν are constants. The line l_1 is $\parallel$ to l_2 and l_1 is $\perp$ to l_3 .

<i>I.</i>	$x = -4,$	$y = 6,$	$z = 10$
<i>II.</i>	$x = 8,$	$y = -12,$	$z = 20$
<i>III.</i>	$x = -4,$	$y = 6,$	$z = -10$

Which of the following is true?

A. *I* and *II*.

B. *I* and *III*

C. *II* and *III*.

D. *I*, *II* and *III*

End of Section 1

Section II

90 marks

Attempt Questions 11 to 16 Allow approximately 2 h 45 min for this section

Write your answers on the writing booklet supplied.

Your responses should include relevant mathematical reasoning and/or calculations.

Write your answers on the writing booklet supplied.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 Marks)

- (a) Solve the equation $z^2 - (1 - 4i)z - 5 + i = 0$ 3

Give the answer in $a + ib$ form where a and b are real

- (b) (i) The complex number z and ω are such that $z = \frac{3a - 5i}{1 + 2i}$ and $\omega = 1 + 13bi$, 2
where a and b are real. Given that $\bar{z} = \omega$, find the exact values of a and b .

- (ii) On an Argand diagram, sketch the regions representing z 2
where $\operatorname{Re}(z + 1) \leq 2$ and $|z - 2i| \leq 2$.

Hence find the greatest value of $|z + 3|$ and the complex number z 3
when $|z + 3|$ is maximum.

- (c) Complex numbers p and q are given by $p = \frac{1 + i}{1 - i}$ and $q = \frac{\sqrt{2}}{1 - i}$

On an Argand diagram with origin O , Sketch the points P , Q and R representing 2
respectively p , q and $p + q$ respectively.

Hence verify that $\arg\left(\frac{1 + \sqrt{2} + i}{1 - i}\right) = \frac{3\pi}{8}$ 3

End of Question 11

Question 12. (15 Marks)

(a) Decompose $\frac{x^2}{4x^2 - 9}$ into $A + \frac{B}{2x - 3} + \frac{C}{2x + 3}$ where A, B and C are real and 4

hence evaluate: $\int_0^1 \frac{x^2}{4x^2 - 9} dx$

(b) Using integration by parts to show that: $\int_1^2 \tan^{-1} \sqrt{x^2 - 1} \, dx = \frac{2\pi}{3} - \ln 3$ 2

(c) Find $\frac{d}{dx} \left(\frac{\ln x}{x} \right)$ 1

Hence find: $\int \frac{1 - \ln x}{x \ln x} dx$ 2

(d) Show that $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{d\theta}{1 + \sec \theta} = \frac{\pi}{12} + \sqrt{2} - 1 - \frac{1}{\sqrt{3}}$ 3

(e) p, q and k are real numbers. If $\int_0^{\frac{\pi}{2}} \sin \left(\frac{p}{p+q} x \right) \cos \left(\frac{q}{p+q} x \right) dx = k$, 3

Find the value of k in terms of p and q .

End of Question 12

Question 13. (15 Marks)

- (a) The position vectors of the points A, B and C are $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $4\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $10\mathbf{i} + 5\mathbf{j} + 9\mathbf{k}$ respectively.

(i) Show that A, B and C are collinear. 2

(ii) Find the exact length of projection of $\overrightarrow{OA}$ on the line OB. 2

(iii) Relative to the origin O, points A and B have position vectors 2

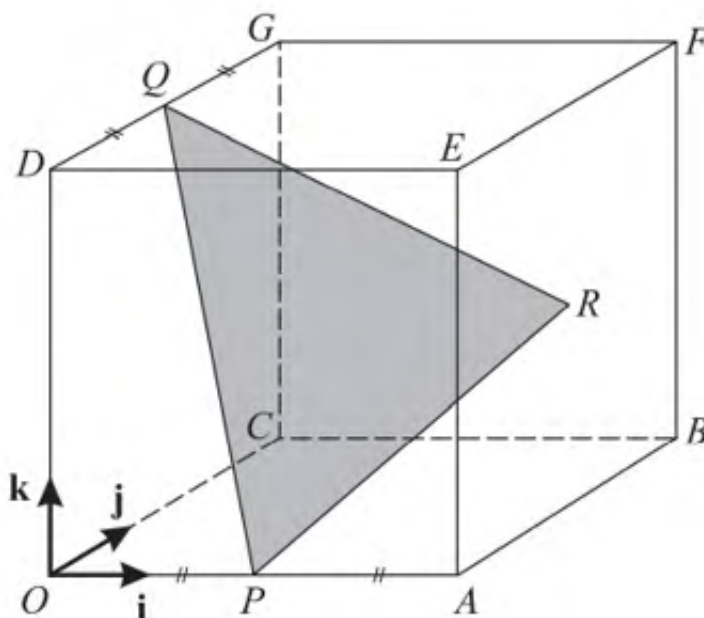
$\begin{bmatrix} c \\ 3 \\ d \end{bmatrix}$ and $\begin{bmatrix} 10 \\ 12 \\ 2 \end{bmatrix}$ respectively, where c and d are constants.

The straight line through A and B has the equation $\mathbf{r} = \begin{bmatrix} 10 \\ 12 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$

where λ is a constant. Find the values of c and d.

Question 13 continues on the next page —

- (b) The diagram below shows a cube $OABCDEFG$ in which the length of each side is 4 units. The unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to $\overrightarrow{OA}$, $\overrightarrow{OC}$ and $\overrightarrow{OD}$ respectively. The mid-points of OA and DG are P and Q respectively and R is the centre of the square face $ABFE$.



- | | |
|--|---|
| (i) Express each of the vectors $\overrightarrow{PR}$, and $\overrightarrow{PQ}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. | 2 |
| (ii) Use scalar product to find angle QPR | 2 |
| (iii) Find the area of triangle PQR correct to 1 decimal place | 2 |
- (c) "If both ab and $a + b$ are even then both a and b are even". 3
 Prove this statement by proving its contrapositive.

End of Question 13

Question 14. (15 Marks)

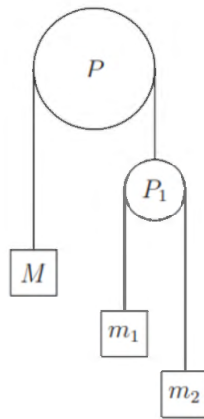
(a) Show that $\frac{(x+k)^2}{x^2+x+1} \leq \frac{4}{3}(k^2-k+1)$ for all real x and k 3

(b) Use the substitution $x = \cos^2 \theta + 4\sin^2 \theta$ to show that :

$$\int_1^4 \frac{x}{\sqrt{(x-1)(4-x)}} dx = \frac{5\pi}{2} \quad 4$$

(c) If $I_n = \int_0^1 x^n (1+x^3)^7 dx$, show that $I_n = \frac{256}{n+22} - \frac{n-2}{n+22} I_{n-3}$ 3

(d) The diagram below show a system of particles, strings and pulleys. In the system, the pulleys are smooth and light, the strings are light and inextensible, the particles move vertically and the pulley labelled with P is fixed. The masses of the particles are as indicated on the diagrams.



For the pulley system shown above, show that the acceleration, a , of the particle of mass M , measured 5

in the downwards direction, is given by $a = \frac{M - 4\mu}{M + 4\mu}g$

where $\mu = \frac{m_1 m_2}{m_1 + m_2}$

End of Question 14

Question 15. (15 Marks)

- (a) In an aerobatics display, Jane and Karen jump from a great height and go through a period of free fall before opening their parachutes. While in free fall at speed v , Jane experiences air resistance kv per unit mass but Karen, who spread-eagles, experience air resistance $\left(kv + \frac{2k^2v^2}{g}\right)$ per unit mass.

(i). Show that Jane's speed can never reach $\frac{g}{k}$ 3

(ii). Obtain the corresponding result for Karen 4

(iii). Jane opens her parachute when her speed is $\frac{g}{3k}$. 2

Show that she has been in free fall for time $\frac{1}{k} \ln \frac{3}{2}$

(iv). Karen also opens her parachute when her speed is $\frac{g}{3k}$. 2

Find the time she has then been in free fall.

(b) P, Q, R and S are real numbers. Show that if $\frac{P}{Q} = \frac{R}{S}$, then $\frac{P}{Q} = \frac{P+R}{Q+S}$ 1

Hence if $\frac{\cos x + \cos y + \cos z}{\cos(x+y+z)} = \frac{\sin x + \sin y + \sin z}{\sin(x+y+z)} = T$, then show that 3

$$T = \cos(x+y) + \cos(y+z) + \cos(z+x)$$

End of Question 15

Question 16. (15 Marks)

(a) Given that $I = \int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \int_0^a \frac{f(a-x)}{f(x) + f(a-x)} dx = \frac{1}{2}a$, 3

evaluate the integral $\int_0^1 \frac{\ln(x+1)}{\ln(2+x-x^2)} dx$

(b) Use a suitable substitution to evaluate $\int_{0.5}^2 \frac{\sin x}{x[\sin x + \sin(\frac{1}{x})]} dx$ 3

(c) The cartesian equations of two lines are given by $y = m_1 + c_1$ and $y = m_2 + c_2$.

(i) find the vector equations of the lines 1

(ii) Hence show that the acute angle of the intersection of the lines is given by 2

$$\tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$$

(d) (i) Let $y = (x-a)^n e^{bx} \sqrt{1+x^2}$, where n and a are constants and b is a non-zero constant. 3

Show that $\frac{dy}{dx} = \frac{(x-a)^{n-1} e^{bx} q(x)}{\sqrt{1+x^2}}$, where $q(x)$ is a cubic polynomial.

(ii) Hence determine 3

$$\int \frac{(x-2)^6 e^{4x} (4x^4 + x^3 - 2)}{\sqrt{1+x^2}} dx$$

—End of Paper—

NSBT Mathematics Ext 2 - Trial 2021 (Task 3)
Sample solutions

1. B
2. D
3. D
4. A
5. B
6. C
7. A
8. A, C
9. B
10. C

$$z - (1-4i)z - 5+i = 0$$

$$z = \frac{(1-4i) \pm \sqrt{(1-4i)^2 + 4(5-i)}}{2}$$

$$= \frac{(1-4i) \pm \sqrt{5-12i}}{2}$$

$$= \frac{(1-4i) \pm (3-2i)}{2}$$

$$= \underline{(2-3i)} \text{ or } \underline{(-1-i)}$$

1b) ii) $z = \frac{3a-5i}{1+2i}$, $w = 1+13bi$

$$\bar{z} = w \Rightarrow \frac{3a+5i}{1-2i} = 1+13bi$$

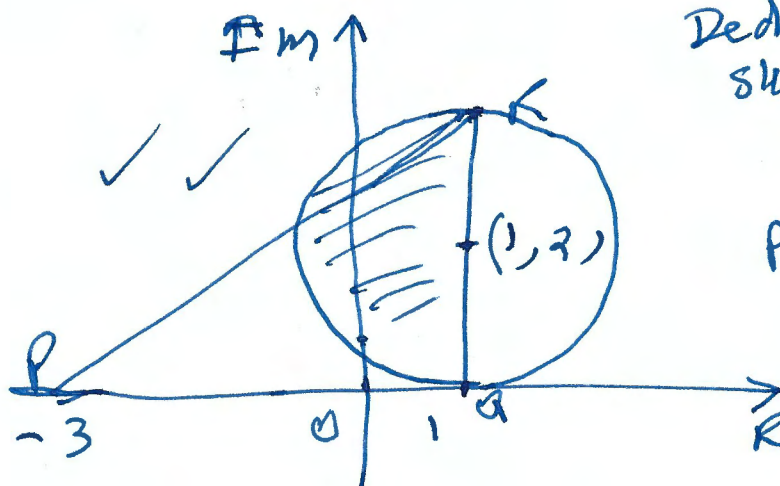
$$\Rightarrow 3a+5i = (1+26b) + (13b-2i)$$

Matching the real and imaginary parts both sides,

$$\left. \begin{array}{l} 26b+1 = 3a \\ \text{and } 13b-2 = 5 \end{array} \right\} \Rightarrow \underline{a=5}, \underline{b=\frac{7}{13}}$$

(iii) $\text{Re}(z+1) \leq 2 \equiv$ region left of the line $\text{Re } z = 1$

$|z-1-2i| \leq 2 \equiv$ circular region centered at $(1, 2)$



Deduce from the sketch that $|z+3|_{\max} = PR$

$$PR = \sqrt{PQ^2 + QR^2}$$

$$= \sqrt{16 + 16}$$

$$= 4\sqrt{2}$$

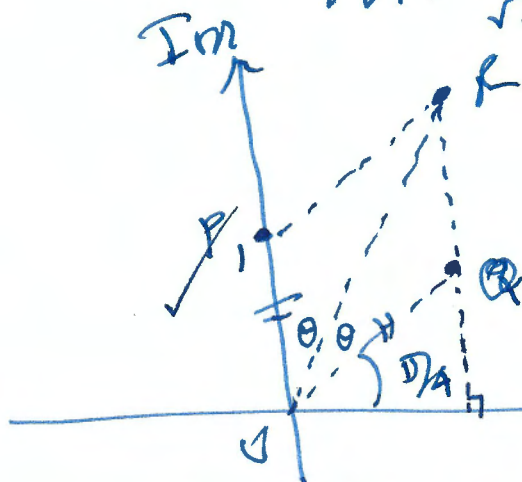
when z is at Q ,
 $z = 1+4i$

$$p = \frac{1+i}{1-i} \Rightarrow \arg p = \frac{\pi}{4} - (-\frac{\pi}{4}) = \frac{\pi}{2}$$

$$|p| = \frac{|1+i|}{|1-i|} = \frac{\sqrt{2}}{\sqrt{2}} = 1$$

$$q = \frac{\sqrt{2}}{1-i} \Rightarrow \arg q = 0 - (-\frac{\pi}{4}) = \frac{\pi}{4}$$

$$|q| = \frac{\sqrt{2}}{\sqrt{2}} = 1$$



$OPQR$ is a rhombus
 $\Rightarrow \angle POR = \angle ROQ = \theta$

Deduce that $\frac{1+\sqrt{2}+i}{1-i} = \frac{1+i}{1-i} + \frac{\sqrt{2}}{1-i} = p+q$ represents
in R

$$\arg\left(\frac{1+\sqrt{2}+i}{1-i}\right) = \angle ROX = \angle ROQ + \angle QOX$$

$$= \theta + \frac{\pi}{4}$$

$$= \frac{1}{2}\left(\frac{\pi}{2} - \frac{\pi}{4}\right) + \frac{\pi}{4}$$

$$= \frac{3\pi}{8}$$

$$\frac{x^2}{4x^2-9} = A + \frac{B}{2x-3} + \frac{C}{2x+3}$$

$$\Rightarrow A = \frac{1}{4}, B = \frac{3}{8}, C = -\frac{3}{8}$$

$$\begin{aligned} \therefore \int_0^1 \frac{x^2}{4x^2-9} dx &= \int_0^1 \frac{1}{4} + \frac{3/8}{2x-3} - \frac{3/8}{2x+3} dx \\ &= \left[\frac{x}{4} + \frac{3}{16} \ln \left| \frac{2x-3}{2x+3} \right| \right]_0^1 \\ &= \frac{1}{4} + \frac{3}{16} \ln \left(\frac{1}{5} \right) \end{aligned}$$

$$(b) \int_1^2 \tan^{-1}(\sqrt{x^2-1}) dx = \left[x \tan^{-1} \sqrt{x^2-1} - \int \frac{1}{\sqrt{x^2-1}} dx \right]_1^2$$

$$\begin{aligned} &= \left(x \tan^{-1} \sqrt{x^2-1} - \ln [x + \sqrt{x^2-1}] \right)_1^2 \\ &= \frac{2\sqrt{3}}{3} - \ln(2 + \sqrt{3}) \end{aligned}$$

$\int_1^2 \frac{1}{\sqrt{x^2-1}} dx$ can be evaluated using the Sub $x = \sec \theta$.

$$(c) \frac{d}{dx} \left(\frac{\ln x}{x} \right) = \frac{1 - \ln x}{x^2}$$

$$\int \frac{1 - \ln x}{x^2} dx = \int \frac{1 - \ln x}{x^2} \cdot \frac{x}{\ln x} dx$$

$$= \int \frac{d}{dx} \left(\frac{\ln x}{x} \right) \cdot \frac{1}{\left(\frac{\ln x}{x} \right)} dx$$

$$= \int \frac{1}{\left(\frac{\ln x}{x} \right)} d \left(\frac{\ln x}{x} \right)$$

$$= \ln \left(\frac{\ln x}{x} \right) + C$$

$$\int_{\pi/4}^{\pi/3} \frac{d\theta}{1 + \sec\theta}$$

$$= \int_{\pi/4}^{\pi/3} \frac{\sec\theta - 1}{\tan^2\theta} d\theta$$

$$= \int_{\pi/4}^{\pi/3} \cot\theta \csc\theta - \cot^2\theta + 1 d\theta$$

$$= \left[-\csc\theta + \cot\theta + \theta \right]_{\pi/4}^{\pi/3}$$

$$= \left(-\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} + \frac{\pi}{3} \right) - \left(-\sqrt{2} + 1 + \frac{\pi}{4} \right)$$

$$= \frac{\pi}{12} + \sqrt{2} - 1 + \frac{1}{\sqrt{3}}$$

$$(e) \int_0^{\pi/2} \sin\left(\frac{p}{p+q}\right)x \cdot \sin\left(\frac{q}{p+q}\right)x dx$$

$$= \frac{1}{2} \int_0^{\pi/2} \cos\left(\frac{p-q}{p+q}\right)x - \cos x dx$$

$$= \frac{1}{2} \left[\frac{p+q}{p-q} \sin\left(\frac{p-q}{p+q}\right)x - \sin x \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\left(\frac{p+q}{p-q} \right) \sin\left(\frac{p-q}{p+q}\right) \cdot \frac{\pi}{2} - 1 \right] = k$$

$$(i) \vec{AB} = (4\hat{i} + 3\hat{j} + 5\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= 3\hat{i} + \hat{j} + 2\hat{k}$$

$$\vec{AC} = (10\hat{i} + 5\hat{j} + 9\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= 9\hat{i} + 3\hat{j} + 6\hat{k}$$

$$= 3(3\hat{i} + \hat{j} + 2\hat{k})$$

$$= 3\vec{AB}$$

$\therefore A, B$ and C are collinear.

(ii) The length of the projection of $\vec{OA}$ onto $\vec{OB}$

$$= \left| \text{proj}_{\vec{OB}} \vec{OA} \right| = |\vec{OA}| \cos \theta$$

$$= \frac{\vec{OA} \cdot \vec{OB}}{|\vec{OB}|}$$

$$= \frac{(4 + 6 + 15)}{\sqrt{4^2 + 3^2 + 5^2}}$$

$$= \frac{25}{\sqrt{50}}$$

$$= \frac{5}{\sqrt{2}}$$

$$(iii) \begin{bmatrix} 10 \\ 12 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} c \\ 3 \\ d \end{bmatrix}$$

$$\Rightarrow \begin{matrix} 10 + \lambda = c \\ 12 + 3\lambda = 3 \\ 2 = d \end{matrix} \Rightarrow \lambda = -3$$

$$\therefore \begin{cases} c = 7 \\ d = 2 \end{cases}$$

$$1/3) \quad \vec{p} = 2\hat{i}$$

$$\vec{q} = 2\hat{i} + 4\hat{k}$$

$$\vec{r} = 4\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{PR} = \vec{r} - \vec{p} = (2\hat{i} + 2\hat{j} + 2\hat{k})$$

$$\vec{PQ} = \vec{q} - \vec{p} = (-2\hat{i} + 2\hat{j} + 4\hat{k})$$

$$(ii) \quad \cos \angle QPR = \frac{\vec{PR} \cdot \vec{PQ}}{|\vec{PR}| \cdot |\vec{PQ}|}$$

$$= \frac{-4 + 4 + 8}{\sqrt{2^2 + 2^2 + 2^2} \cdot \sqrt{2^2 + 2^2 + 4^2}}$$

$$= \frac{8}{12\sqrt{2}}$$

$$= \frac{\sqrt{2}}{3}$$

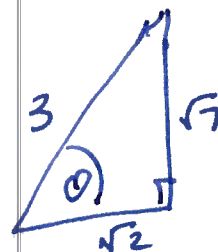
$$\therefore \angle QPR =$$

$$(iii) \quad \sin \angle QPR = \frac{\sqrt{7}}{3}$$

$$\text{Area of } \triangle PQR = \frac{1}{2} |\vec{PR}| |\vec{PQ}| \sin \angle QPR$$

$$= \frac{1}{2} \sqrt{12} \sqrt{24} \cdot \frac{\sqrt{7}}{3}$$

$$= 2\sqrt{14} \text{ sq. units}$$



13 (c)

We can prove the statement by proving its contrapositive:

"If a and b are ^{not} both even, then ab and $(a+b)$ are not both even"

(1c) If ~~a~~ either a or b is odd; then either ab or $a+b$ is odd

Suppose a is odd. Then $a = 2k$ for some integer k .

Then $ab = (2k+1)b = 2kb + b$ which is odd if b is odd; even otherwise

And $a+b = 2k+1+b$ which is odd if L is even, and even otherwise.

So regardless of whether L is even or odd, (it must be one or the other), one of (ab) or $(a+b)$ is necessarily odd.

The argument is identical if we happen to be odd.

$$1 (a) \text{ let } y = \frac{(x+k)^2}{x^2+x+1}$$

Rearranging it gives, ✓

$$(y-1)x^2 + (y-2k)x + (y-k^2) = 0$$

This is a quadratic equation in x .

$$\therefore \forall \text{ real } x, \Delta \geq 0$$

$$\Rightarrow (y-2k)^2 - 4(y-1)(y-k^2) \geq 0$$

$$\Rightarrow 3y^2 - 4(k^2 - k + 1)y \leq 0$$

$$\Rightarrow y \leq \frac{4}{3}(k^2 - k + 1)$$

$$(b) \quad x = \cos^2 \theta + 4 \sin^2 \theta$$

$$\frac{dx}{d\theta} = -2 \cos \theta \sin \theta + 8 \sin \theta \cos \theta$$

$$= 3 \sin 2\theta$$

$$x=1 \quad 1 = \cos^2 \theta + 4 \sin^2 \theta \Rightarrow \sin^2 \theta = 0 \Rightarrow \theta = 0$$

$$x=4 \quad 4 = \cos^2 \theta + 4 \sin^2 \theta \Rightarrow \cos^2 \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

$$\Rightarrow \int_1^4 \frac{x}{\sqrt{(x-1)(4-x)}} dx = \int_0^{\pi/2} \frac{\cos^2 \theta + 4 \sin^2 \theta}{\sqrt{3 \sin^2 \theta \cdot 3 \cos^2 \theta}} \cdot 3 \sin 2\theta d\theta$$

$$= \int_0^{\pi/2} 2 \cos^2 \theta + 8 \sin^2 \theta d\theta$$

$$= \int_0^{\pi/2} (5 - 3 \cos 2\theta) d\theta$$

$$= \left[5\theta - \frac{3}{2} \sin 2\theta \right]_0^{\pi/2}$$

$$= \frac{5\pi}{2}$$

14(c)

$$I_n = \int_0^1 x^n (1+x^3)^7 dx$$

$$= \int_0^1 \frac{x^{n-2}}{3} \cdot 3x^2 (1+x^3)^7 dx$$

$$= \left[\frac{x^{n-2} (1+x^3)^8}{24} - \int \frac{(1+x^3)^8}{24} \cdot (n-2)x^{n-3} dx \right]_0^1$$

$$= \frac{256}{24} - \frac{(n-2)}{24} \int_0^1 x^{n-3} (1+x^3)^8 dx$$

$$= \frac{256}{24} - \frac{(n-2)}{24} \int_0^1 x^{n-3} (1+x^3)^7 (1+x^3) dx$$

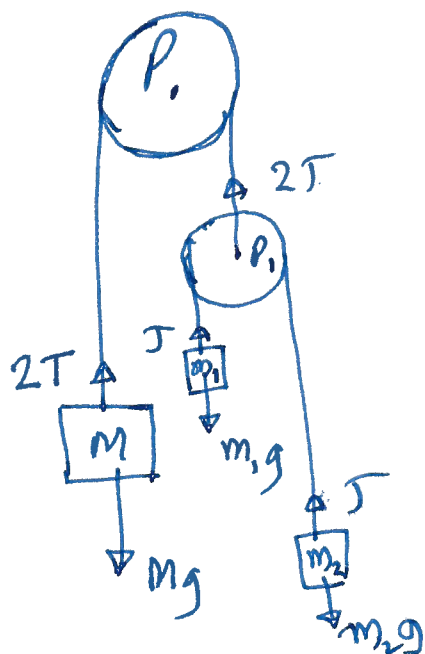
$$= \frac{256}{24} - \frac{(n-2)}{24} \left[\int_0^1 x^{n-3} (1+x^3)^7 + x^n (1+x^3)^7 dx \right]$$

$$= \frac{256}{24} - \frac{(n-2)}{24} [I_{n-3} + I_n]$$

$$I_n = \frac{256}{24} - \frac{(n-2)}{24} [I_{n-3} + I_n]$$

$$I_n \left[1 + \frac{n-2}{24} \right] = \frac{256}{24} - \frac{(n-2)}{24} I_{n-3}$$

$$\Rightarrow I_n = \frac{256}{(n+22)} - \frac{(n-2)}{(n+22)} I_{n-3}$$



Let M moves downward with acceleration a and m_2 moves downward with respect to P_1 with acceleration a_1 .

Applying Newton's 2nd law on:

$$\underline{M} \quad \downarrow \quad Mg - 2T = Ma \quad \text{--- (1) ✓}$$

$$\underline{m_1} \quad \uparrow \quad T - m_1g = m_1(a + a_1) \quad \text{--- (2) ✓}$$

$$\underline{m_2} \quad \uparrow \quad T - m_2g = m_2(a - a_1) \quad \text{--- (3) ✓}$$

$$m_2 \times (2) + m_1 \times (3)$$

$$\Rightarrow (m_1 + m_2)T - 2m_1m_2g = 2m_1m_2a \quad \text{--- (4)}$$

$$(m_1 + m_2) \times (1) + 2 \times (4)$$

$$M(m_1 + m_2)g - 4m_1m_2g = [M(m_1 + m_2) + 2m_1m_2]a$$

$$\therefore a = \frac{[M(m_1 + m_2) - 4m_1m_2]g}{M(m_1 + m_2) + 2m_1m_2}$$

Dividing by all by $(m_1 + m_2)$

$$a = \frac{\left[M - 4 \frac{m_1m_2}{m_1 + m_2}\right]g}{\left[M + 2 \frac{m_1m_2}{m_1 + m_2}\right]}$$

$$= \frac{(M - 4n)g}{m + 2n}$$

Where $n = \frac{m_1m_2}{m_1 + m_2}$

17 June

Applying $F = ma$

$$\downarrow mg - mkv = m\ddot{x}$$

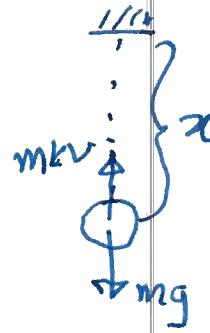
$$\ddot{x} = g - kv \quad \checkmark$$

$$\frac{dv}{dt} = g - kv$$

$$\int_0^v \frac{dv}{g - kv} = \int_0^t dt \quad \checkmark$$

$$-\frac{1}{k} \ln\left(1 - \frac{kv}{g}\right) = t$$

For finite values of t , $\frac{kv}{g} - 1 \neq 0$
 $\Rightarrow v \neq \frac{g}{k}$



(ii) Karen

Applying $F = ma$

$$\downarrow mg - m\left(kv + 2k\frac{v^2}{g}\right) = m\ddot{x}$$

$$\ddot{x} = g - kv - 2k\frac{v^2}{g} \quad \checkmark$$

$$= -\frac{1}{g} \left[2k^2 v^2 + kv - g^2 \right]$$

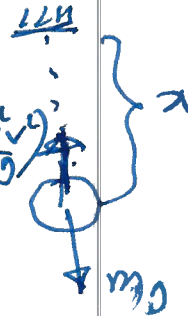
$$\ddot{x} = -\frac{1}{g} \left[(2kv - g)(kv + g) \right]$$

$$\int_0^v \frac{g dv}{(2kv - g)(kv + g)} = \int_0^t dt \quad \checkmark$$

$$\int_0^v \left(\frac{-2/3}{2kv - g} + \frac{1/3}{g + kv} \right) dv = \int_0^t dt$$

$$\frac{1}{3k} \ln \left| \frac{g + kv}{2kv - g} \right| = t$$

for finite values of t ,
 $2kv - g \neq 0$
 $\Rightarrow v \neq \frac{g}{2k}$



15 (a) (iii) When Jane's speed = $\frac{g}{3k}$,

$$-\frac{1}{k} \ln\left(1 - \frac{1}{3}\right) = t$$

$$t = -\frac{1}{k} \ln\left(\frac{2}{3}\right)$$

$$\Rightarrow t = \frac{1}{k} \ln\left(\frac{3}{2}\right)$$

$\therefore$ Jane has been in free fall for $\frac{1}{k} \ln\left(\frac{3}{2}\right)$ seconds

15 (a) (iv) When Karen's speed = $\frac{g}{3k}$,

$$\frac{1}{3k} \ln\left|\frac{4/3}{1/3}\right| = t$$

$$\Rightarrow t = \frac{1}{3k} \ln 4$$

$\therefore$ Karen has been in free fall for $\frac{1}{3k} \ln 4$ seconds

15 (b) (i) Let $\frac{P}{Q} = \frac{iR}{iS} = K$

$$\Rightarrow P = QK$$

$$iR = iSK$$

$$P + iR = (Q + iS)K$$

$$\Rightarrow K = \frac{P + iR}{Q + iS}$$

$$\Rightarrow \frac{P}{Q} = \frac{P + iR}{Q + iS}$$

16/1/21

$$\frac{P}{Q} = \frac{R}{S} = \frac{P+iR}{Q+iS}$$

$$= \frac{iR}{iS} = \frac{P+iR}{Q+iS}$$

Let $P = \cos x + \cos y + \cos z$

$Q = \cos(x+y+z)$

$R = \sin x + \sin y + \sin z$ ✓

$S = \sin(x+y+z)$

~~$\frac{P}{Q} = \frac{iR}{iS}$~~

Correct $\frac{P}{Q} = \frac{P+iR}{Q+iS} = T$ ✓

$$\Rightarrow T = \frac{(\cos x + \cos y + \cos z) + i(\sin x + \sin y + \sin z)}{\cos(x+y+z) + i \sin(x+y+z)}$$

$$= \frac{e^{ix} + e^{iy} + e^{iz}}{e^{i(x+y+z)}}$$

$$= e^{-i(y+x)} + e^{-i(z+x)} + e^{-i(y+z)}$$

But T is Real

$$\Rightarrow T = \text{Re} \left[e^{-i(y+x)} + e^{-i(y+z)} + e^{-i(z+x)} \right]$$

$$= \cos(x+y) + \cos(y+z) + \cos(z+x)$$

$$1b/ (0) \int_0^1 \frac{\ln(x+1)}{\ln(2+x-x^2)} dx$$

$$= \int_0^1 \frac{\ln(x+1)}{\ln[(x+1)(-x+2)]} dx \checkmark$$

$$= \int_0^1 \frac{\ln(x+1)}{\ln(x+1) + \ln(2-x)} dx$$

$$= \int_0^1 \frac{\ln(x+1)}{\ln(x+1) + \ln[1-(x-1)]} dx \checkmark$$

$$\begin{bmatrix} f(x) = \ln(x+1) \\ f(1-x) = \ln(1-x+1) = \ln(2-x) \end{bmatrix}$$

$$= \frac{1}{2} \checkmark$$

$$(b) \text{ let } I = \int_{0.5}^2 \frac{\sin x}{x [\sin x + \sin(\frac{1}{x})]} dx$$

$$\text{let } u = \frac{1}{x} \Rightarrow du = -\frac{1}{x^2} dx \checkmark$$

$$x = 0.5, u = 2$$

$$x = 2, u = 0.5$$

$$\Rightarrow I = \int_2^{0.5} \frac{\sin(\frac{1}{u})}{(\cancel{x}) [\sin(\frac{1}{u}) + \sin(u)]} \cdot -x^2 du$$

$$= \int_{0.5}^2 \frac{\sin(\frac{1}{u}) du}{u [\sin(\frac{1}{u}) + \sin(u)]} \checkmark$$

$$\Rightarrow 2I = \int_{0.5}^2 \frac{1}{x} dx = \int_{0.5}^2 \frac{\sin(\frac{1}{x})}{x [\sin x + \sin(\frac{1}{x})]} dx$$

$$\Rightarrow 2I = \int_{0.5}^2 \frac{1}{x} dx = [\ln x]_{0.5}^2 = \ln 4 = 2 \ln 2 \checkmark$$

Q16

(i) $y = m_1x + c_1$ It passes through $(0, c_1)$ and gradient m_1 ,

$\therefore$ The vector equation is

$$\underline{r} = c_1 \underline{j} + \lambda (\underline{i} + m_1 \underline{j}) \quad (\lambda \in \mathbb{R})$$

✓ Similarly the vector equation of $y = m_2x + c_2$ is

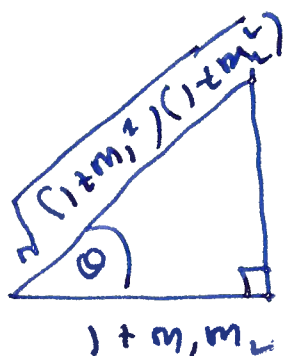
$$\underline{r} = c_2 \underline{j} + \mu (\underline{i} + m_2 \underline{j}) \quad (\mu \in \mathbb{R})$$

(ii) If θ is the acute angle between them

$$\text{Then } (\underline{i} + m_1 \underline{j}) \cdot (\underline{i} + m_2 \underline{j}) = |\underline{i} + m_1 \underline{j}| |\underline{i} + m_2 \underline{j}| \cos \theta$$

$$\Rightarrow 1 + m_1 m_2 = \sqrt{1 + m_1^2} \sqrt{1 + m_2^2} \cdot \cos \theta \quad \checkmark$$

$$\cos \theta = \frac{1 + m_1 m_2}{\sqrt{1 + m_1^2} \sqrt{1 + m_2^2}}$$



$$\tan \theta = \frac{\sqrt{(1 + m_1^2)(1 + m_2^2)} - (1 + m_1 m_2)}{1 + m_1 m_2}$$

$$= \frac{\sqrt{m_1^2 - 2m_1 m_2 + m_2^2}}{1 + m_1 m_2}$$

$$= \frac{m_1 - m_2}{1 + m_1 m_2} \quad \checkmark$$

16 (i) $y = (x-a)^n e^{bx} \sqrt{1+x^2}$

16 (ii) $\frac{dy}{dx} = n(x-a)^{n-1} e^{bx} \sqrt{1+x^2} + (x-a)^n \left[b e^{bx} \sqrt{1+x^2} + e^{bx} \frac{x}{\sqrt{1+x^2}} \right]$

$$= (x-a)^{n-1} e^{bx} \left[n \sqrt{1+x^2} + (x-a) \frac{b(1+x^2) + x}{\sqrt{1+x^2}} \right]$$

$$= \frac{(x-a)^{n-1} e^{bx}}{\sqrt{1+x^2}} \left[n(1+x^2) + b(x-a)(1+x^2) + x(x-a) \right]$$

$$= \frac{(x-a)^{n-1} e^{bx}}{\sqrt{1+x^2}} \left[bx^3 + (n-ab+x) x^2 + (b-a)x + (n-ab) \right]$$

$$= \frac{(x-a)^{n-1} e^{bx}}{\sqrt{1+x^2}} \cdot q(x) \text{ where } q(x) \text{ is a cubic polynomial}$$

(ii) $n=8, a=2, b=4$

① $\int \frac{(x-2)^7 e^{4x}}{\sqrt{1+x^2}} (4x^3 + x^2 + 2x) dx = (x-2)^8 e^{4x} \sqrt{1+x^2}$

② $\int \frac{(x-2)^6 e^{4x}}{\sqrt{1+x^2}} (4x^3 + 2x + 1) dx = (x-2)^7 e^{4x} \sqrt{1+x^2}$

① + 2 × ②

$$\int \frac{(x-2)^6 e^{4x}}{\sqrt{1+x^2}} \left[(x-2)(4x^3 + x^2 + 2x) + 2(4x^3 + 2x + 1) \right] dx$$

$$= (x-2)^7 e^{4x} \sqrt{1+x^2} [(x-2) + 2]$$

$$\Rightarrow \int \frac{(x-2)^6 e^{4x}}{\sqrt{1+x^2}} [4x^4 + x^3 - 2] dx = \frac{x(x-2)^7 e^{4x} \sqrt{1+x^2}}{\sqrt{1+x^2}} + C$$